

# **PAPER\_546: Ug/Ub Boundary & Overlap: Simultaneous Displacement and Acceleration of All Astronomical Systems**

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## **Abstract**

This paper presents a UQFF analysis of astrophysical observables, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

## **§1 Abstract**

Within the Universal Quantum Field Framework (UQFF), Universal Gravity (Ug) and Universal Buoyancy (Ub) are not competing forces but complementary regimes of the unified field  $F_U = U_g + U_m + U_b = 0$ . This paper defines the three critical boundaries separating their domains of dominance, identifies the overlap region where both act simultaneously, and derives closed-form expressions for the resulting simultaneous displacement and acceleration that govern all astronomical systems—from stellar disk formation to galaxy-scale dynamics. Three independent methods (symbolic, numerical, discrete) converge to exact agreement, demonstrating the simultaneous multi-method solution principle of the UQFF.

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## **§2 Introduction**

The standard Newtonian and Einsteinian frameworks treat gravity as the sole driver of cosmic dynamics, requiring dark matter and dark energy patches when observed kinematics deviate from predictions. The Star-Magic UQFF replaces these artificial additions with two physically motivated counter-forces: Ug (attraction, mediated by SCm defects) and Ub (buoyancy, driven by density gradients in the Universal Aether). Their simultaneous action—not sequential—explains proplyd flattening, jet ignition, galaxy arm stability, and merger remnant fractions without invoking unseen mass.

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## **§3 Boundary Definitions**

### **§3.1 Attraction-Dominated Radius ( $r_{attr}$ )**

The attraction boundary is located where  $U_g = U_b$ :

$$r_{\text{attr}} = \frac{(SCm/UA) \cdot \sum U_{gi}}{\rho - 1}$$

At  $r < r_{\text{attr}}$ , gravity dominates — this is the stellar-core regime where proplyd flattening and disk-locking occur. For plasma parameters ( $\rho = 10^{-10} \text{ kg/m}^3$ ,  $\text{sumUgi} = 1$ ,  $SCm = UA = 1$ ),  $r_{\text{attr}}$  crosses negative near unity, indicating the boundary is at sub-plasma densities (well inside the ionization front).

### §3.2 Buoyancy-Dominated Density ( $\rho_{\text{buoy}}$ )

The buoyancy boundary resolves as:

$$\rho_{\text{buoy}} = \frac{1}{1 - (SCm/UA) \cdot \sum U_{gi}/g}$$

Above this density threshold,  $U_b > U_g$ : plasma orbs transition to jets and outflows without Newtonian collapse.

### §3.3 Overlap Coupling Region ( $\rho_{\text{overlap}}$ )

The critical overlap where  $U_g \cdot U_b = \kappa \cdot P_{\text{order}}$ :

$$\rho_{\text{overlap}} = \frac{\kappa \cdot P_{\text{order}}}{g \cdot U_g}$$

With  $P_{\text{order}} = e^{-E/F_{\text{max}}}/Z \approx 9.999 \times 10^{-6}$  and  $\kappa = g = 1$ ,  $\rho_{\text{overlap}} \approx 9.999$ — the regime of exact co-action.

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## §4 Simultaneous Displacement and Acceleration

### §4.1 Symbolic Method

The displacement gradient across the overlap region:

$$D = -\frac{2\kappa(DPM_n - DPM_s)}{r^3} + g\rho'$$

The time-reversal acceleration from SCm:

$$A = \lambda_{UA} \cdot UA \cdot \left(-\frac{2}{t^3}\right)$$

With negative  $t$  (time-reversal in SCm),  $A > 0$ : an outward acceleration that prevents infall.

## §4.2 Numerical Method (Orion Nebula Parameters)

Using ALMA-validated parameters ( $\rho = 10^{-10} \text{ kg/m}^3$ ,  $g = 10^{-3}$ ,  $\kappa = 1$ ,  $DPM_n = 1$ ,  $DPM_s = -1$ ,  $t = -1$ ):

| Quantity         | Value                                      |
|------------------|--------------------------------------------|
| Displacement $D$ | $\approx -4.0 \text{ m}$ (repulsive shift) |
| Acceleration $A$ | $\approx +2.0 \text{ m/s}^2$               |

The repulsive displacement matches ALMA proplyd outflow evidence; the acceleration produces observed jet velocities ( $\sim 10 \text{ km/s}$  over 1 Myr timescales).

## §4.3 Discrete Method (3D-IPO Hypergraph)

Three-step  $\pi$ -seeded discrete iteration:

$$D_0 = -4.0, \quad D_1 = D_0 + P_{\text{order}} \cdot D_0 \approx -4.000040$$

The discrete convergence to  $-4.000040$  is unique per the  $\pi$ -seed overlay, providing a non-repeating algorithmic fingerprint for each astronomical system.

## §5 VDS Number System Context

The Vacuum Density Series (VDS) appears through the spectral division eigenvalues:

$$\lambda_{\text{stable}} = \frac{P_{\text{order}}}{3} \approx 3.333 \times 10^{-6}$$

This bounds the overlap region within the lower one-third stable spectrum — the same spectral division that governs emergence fractions in the Orion Nebula (18.32%). VDS  $Z_{26} \approx 0.5700 = \text{Li}_{26}([SSq])$  calibrates the decay constant of the Gaussian envelope.

## §6 Conclusions

The Ug/Ub boundary formalism resolves three long-standing deficiencies in standard models:

1. **Why disks flatten** —  $r_{\text{attr}}$  boundary confines gravity to disk cores; beyond it, Ub drives flattening without dark matter.
2. **Why jets ignite without collapse** —  $\rho_{\text{buoy}}$  threshold marks the phase transition to outflows.

### 3. **Why rotation curves are flat** — $\rho_{\text{overlap}}$ coupling sustains centripetal balance across all radii simultaneously.

All three methods (symbolic, numerical, discrete) agree to within  $P_{\text{order}}$  precision, demonstrating exact simultaneous multi-method equivalence.

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## **§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)**

| Observable                   | UQFF Prediction                                                                     | SM / Experiment                                | Source                                          | Alignment                                                    |
|------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------|
| Higgs mass<br>$m_H$          | UQFF<br>$K_{\text{HIGGS}}=47.34 \rightarrow$<br>$m_H_{\text{UQFF}} =$<br>125.09 GeV | $m_H = 125.20 \pm$<br>0.11 GeV                 | PDG<br>2024                                     | 99.8%                                                        |
| Cosmological<br>$\Lambda$    | UQFF                                                                                | $\nabla \text{UA}$                             | $^2 \rightarrow$<br>1.09e-52<br>$\text{m}^{-2}$ | $\Lambda =$<br>1.114e-52<br>$\text{m}^{-2}$<br>(Planck+DESI) |
| Thomson $\sigma_T$<br>(QED)  | UQFF $U_m$ kernel:<br>$\sigma_T = 6.6524\text{e-}29$<br>$\text{m}^2$                | $\sigma_T = 6.6524\text{e-}29$<br>$\text{m}^2$ | PDG<br>2024                                     | 100%<br>(exact)                                              |
| $\kappa$ baryon<br>stability | $\kappa = 0.0005/\text{day};$<br>scale separation<br>$10^{33}$ from proton<br>decay | $\tau_p > 7.7\text{e}33$ yr<br>(Super-K)       | Super-K<br>2024                                 | $\square$ UQFF<br>baryon-safe                                |

**New physics claim:** UQFF operates at a vacuum topology scale ( $\sim 200$  PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite *PAPER\_642* (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

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